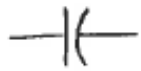
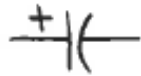


Basic Components and Electric Circuits

PASSIVE COMPONENTS:



FIXED
CAPACITOR



FIXED
CAPACITOR
(POLARIZED)



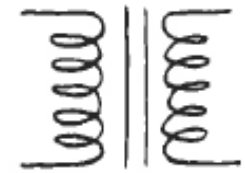
VARIABLE
CAPACITOR



FIXED
RESISTOR



VARIABLE
RESISTOR

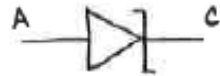


TRANSFORMER

DIODES AND THYRISTORS:



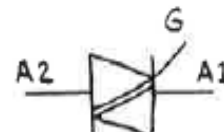
RECTIFIER/
DIODE



ZENER
DIODE



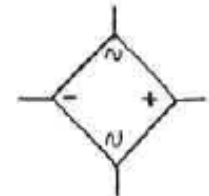
SCR



TRIAC

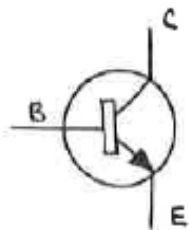


TRIGGER
DIODE

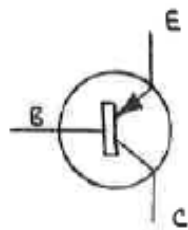


BRIDGE
RECTIFIER

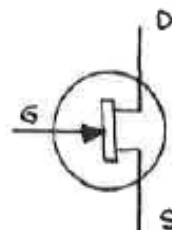
TRANSISTORS:



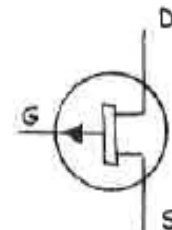
NPN



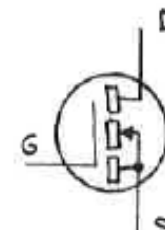
PNP



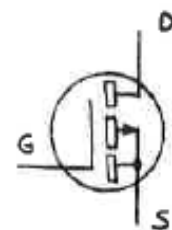
N-JFET



P-JFET



N-MOSFET



P-MOSFET

PHOTONIC COMPONENTS:



LED



PHOTODIODE

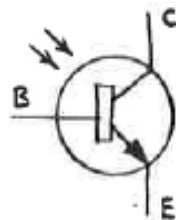
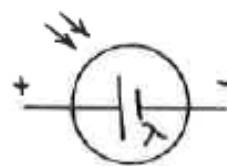
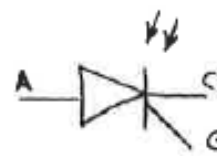


PHOTO-
TRANSISTOR



SOLAR CELL



LASCR

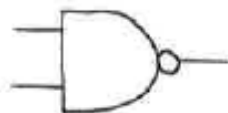


PHOTO-
RESISTOR

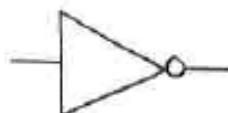
LOGIC GATES:



AND



NAND



INVERTER



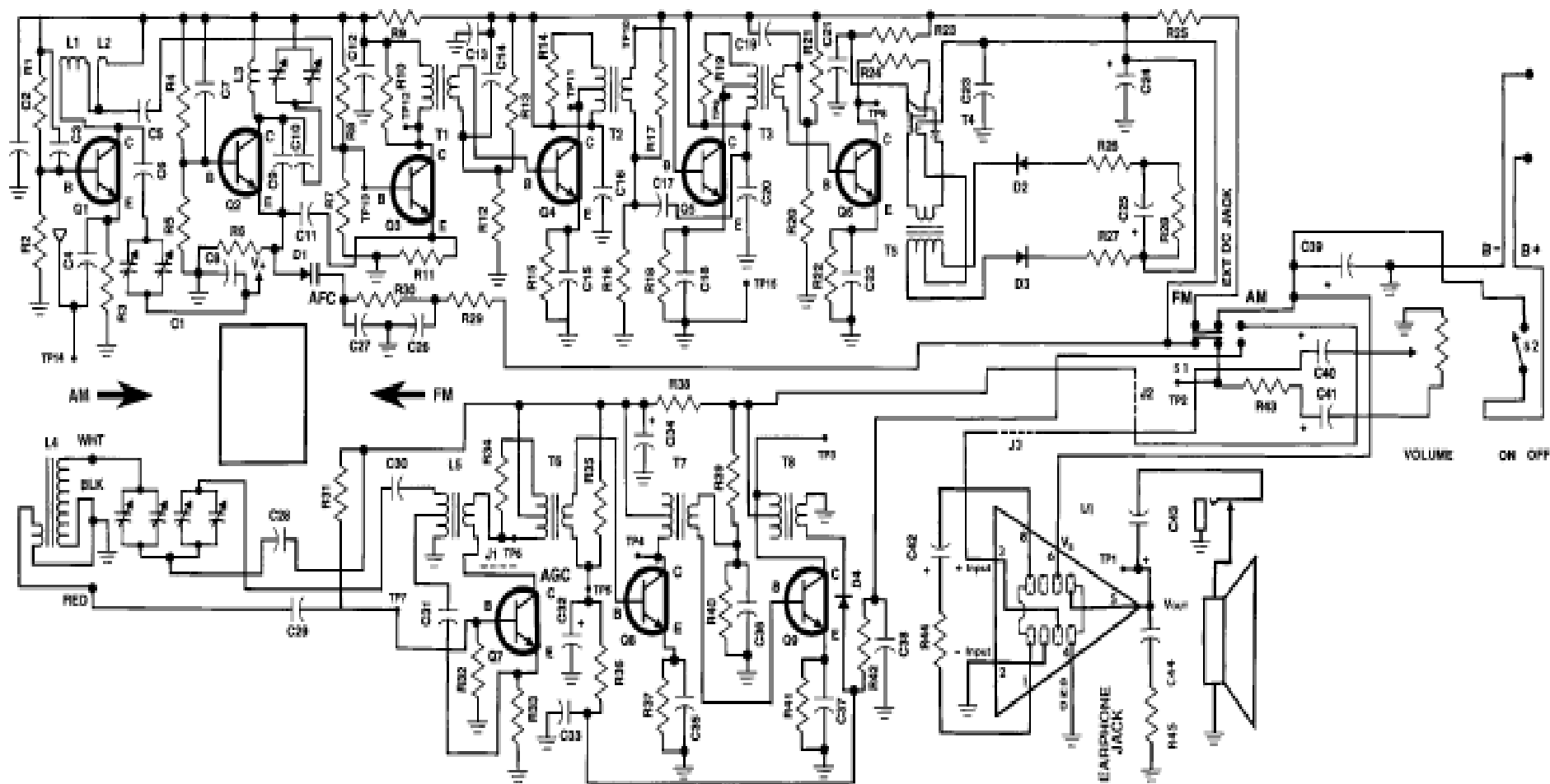
OR

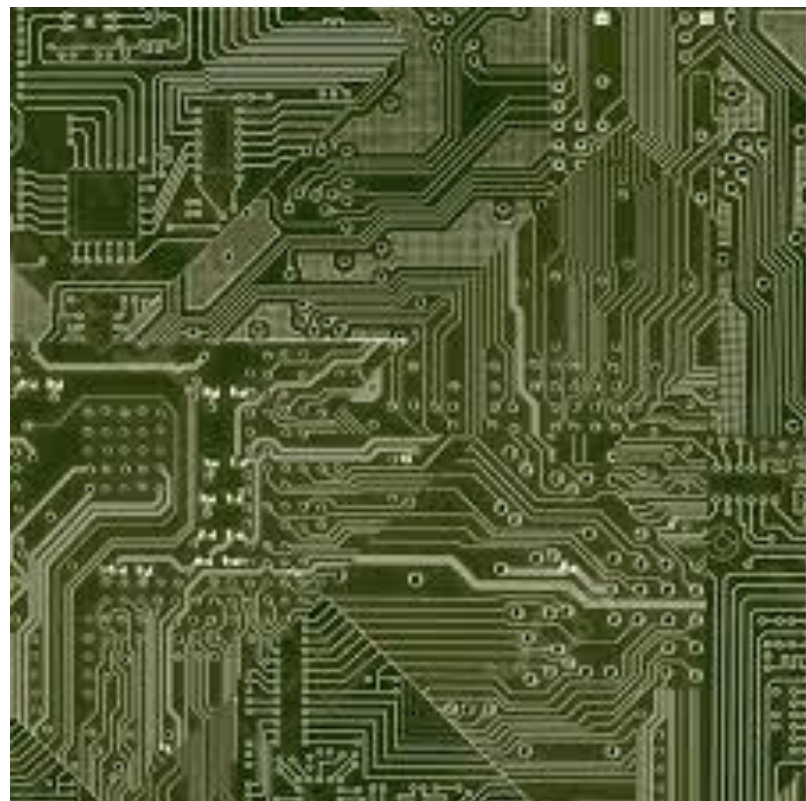


NOR

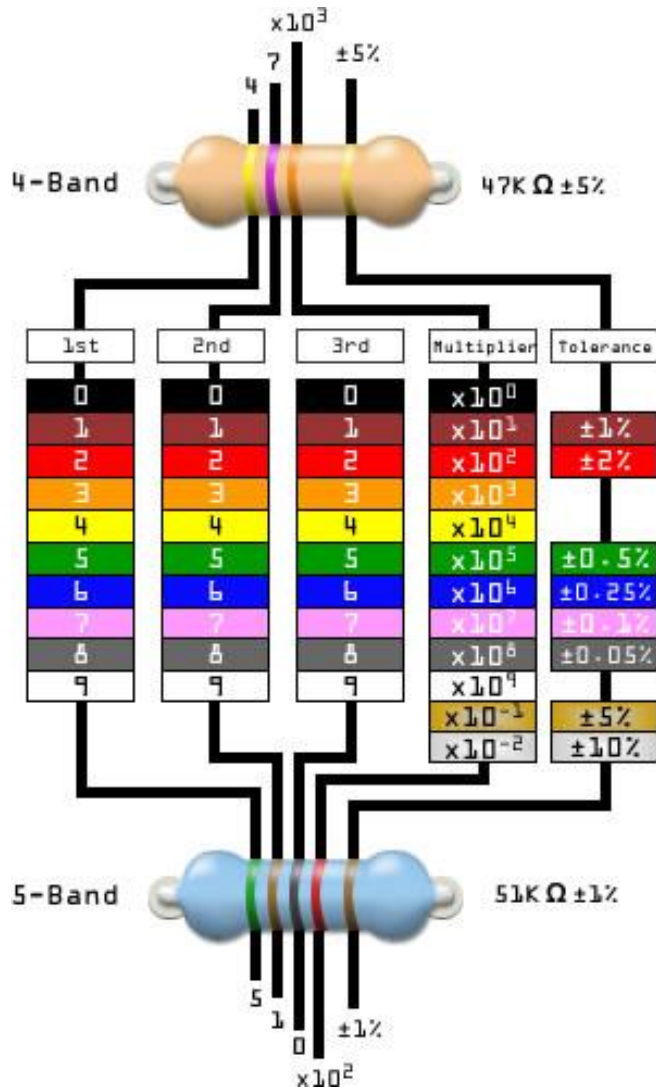


EX-OR





The Resistor Color Code



Select the *first 3 or 4 bands* for 20%, 10% or 5% resistors and *all 5 bands* for precision (2% or less), 5-band resistors.

A few resistors have an additional band indicating either the **reliability** or the **temperature coefficient**. The reliability band specifies the failure rate per 1000 hours (assuming that a full wattage being applied to the resistor). This stripe is found primarily on 4-band resistors made for military applications and seldom used in commercial electronics.

The temperature coefficient is more commonly marked, especially on quality 5-band resistors, as it starts to become an important factor for precision components. For a resistor with temperature coefficient of 200 ppm, for example, a change in temperature of 50°C causes a value change of 1%.

Resistors with tolerance $> 20\%$ can have only 3 bands!

TABLE 1.1 RETMA color code for resistors

Color	1 First Significant Digit	2 Second Significant Digit	3 Multiplier	4 Percent Tolerance
Silver			0.01	10
Gold			0.1	5
Black	(not used)	0	1.0	
Brown	1	1	10	1
Red	2	2	100	2
Orange	3	3	1000	3
Yellow	4	4	10,000	4
Green	5	5	100,000	
Blue	6	6	1,000,000	
Purple	7	7	10,000,000	
Gray	8	8	100,000,000	
White	9	9	1,000,000,000	
No color				20

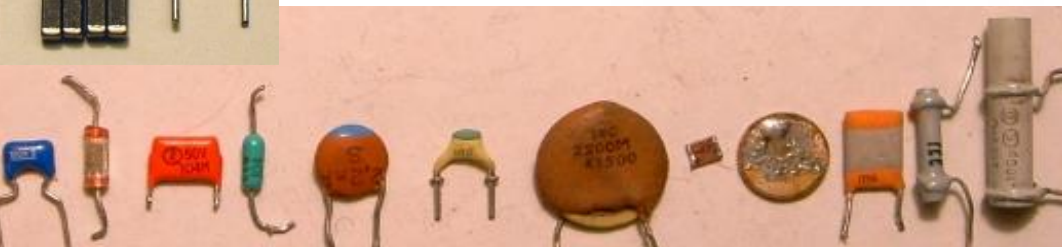
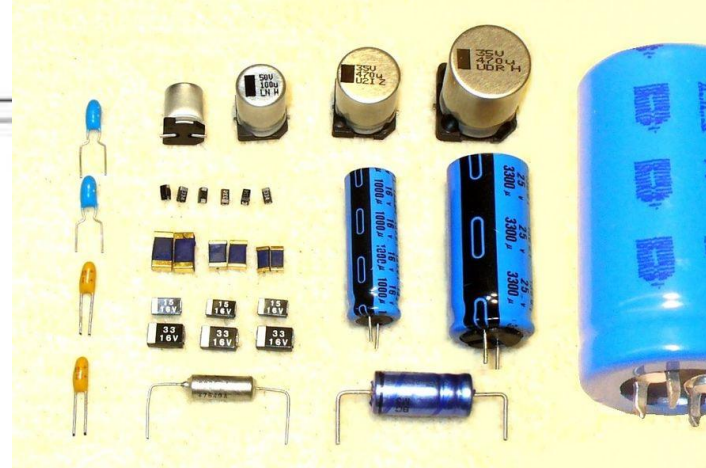
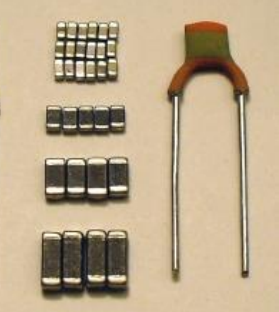


TABLE II.1 Color code for molded ceramic capacitors

1% = 10000

Color	A First Significant Digit	B Second Significant Digit	C Multiplier	D TOLERANCE		G Temperature Coefficient (parts per million per °C)
				Value Greater Than 10 pF (%)	Value 10 pF or Less (pF)	
Black	(not used)	0	1	20	2.0	0
Brown	1	1	10	1		-30
Red	1	1	100	2		-80
Orange	3	3	1000			-150
Yellow	4	4	10,000			-220
Green	5	5	100,000	5	0.5	-330
Blue	6	6	1,000,000			-470
Violet	7	7	10,000,000			-750
Gray	8	8	0.01		0.25	+30

SI Base Units & Prefixes

SI Base Units

Quantity	Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Temperature	kelvin	K
Time	second	s
Amount of Substance	mole	mol
Electric Current	ampere	A
Luminous Intensity	candela	cd

$$1 \text{ J} = 1 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$$

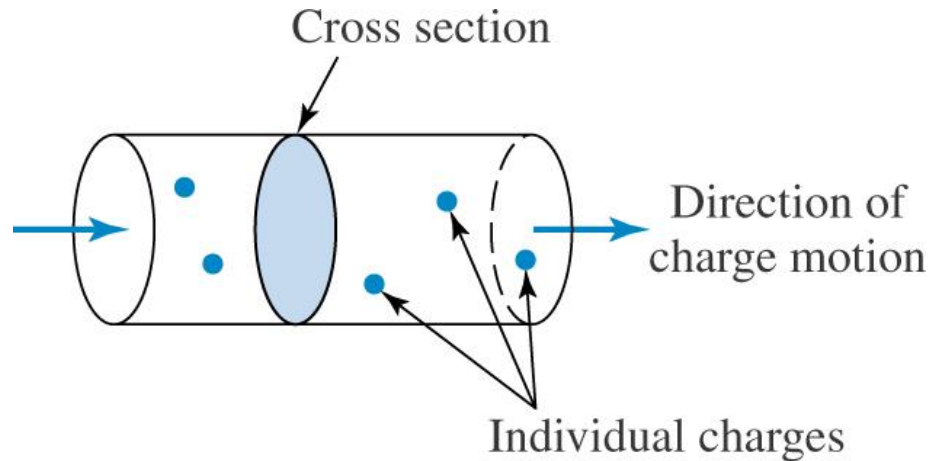
$$1 \text{ W} = 1 \text{ J/s}$$

$$1 \text{ kWh} = 3.6 \cdot 10^6 \text{ J}$$

Prefiks	Symbol	Multiplying factor
yotta	Y	1 000 000 000 000 000 000 000 000 = 10^{24}
zetta	Z	1 000 000 000 000 000 000 000 = 10^{21}
exa	E	1 000 000 000 000 000 000 = 10^{18}
peta	P	1 000 000 000 000 000 = 10^{15}
tera	T	1 000 000 000 000 = 10^{12}
giga	G	1 000 000 000 = 10^9
mega	M	1 000 000 = 10^6
kilo	k	1 000 = 10^3
hecto	h	100 = 10^2
deka	da	10 = 10^1
deci	d	0,1 = 10^{-1}
centi	c	0,01 = 10^{-2}
milli	m	0,001 = 10^{-3}
mikro	μ	0,000 001 = 10^{-6}
nano	n	0,000 000 001 = 10^{-9}
piko	p	0,000 000 000 001 = 10^{-12}
femto	f	0,000 000 000 000 001 = 10^{-15}
atto	a	0,000 000 000 000 000 001 = 10^{-18}
zepto	z	0,000 000 000 000 000 000 001 = 10^{-21}
yocto	y	0,000 000 000 000 000 000 000 001 = 10^{-24}

Charge

- 2 types of charges:
- *positive (protons)*
 - *negative (electrons)*



We neither create nor destroy electrons (or protons) when running electric circuits – **charge conservation**

Charge in motion represents **current**

Unit of charge is **Coulomb** – charge transported by steady current of one **Ampere** in one second: $1\text{C} = 1\text{A}\cdot 1\text{s}$

Charge of **electron** is $-1.602 \times 10^{-19} \text{ C}$

Charge of **proton** is $+1.602 \times 10^{-19} \text{ C}$

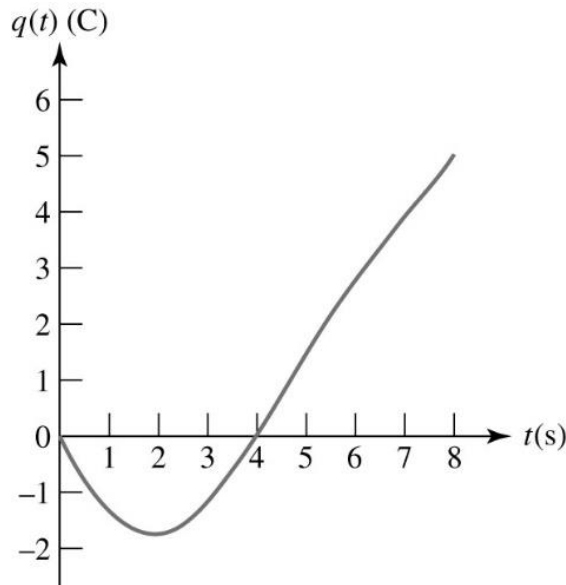
Current

Q – quantity of charge that does not change with time

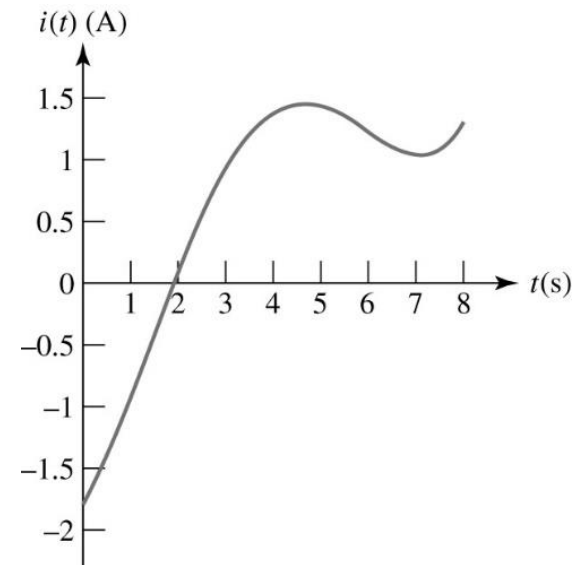
q – quantity of charge that changes over time or instantaneous amount of charge

Charge motion → transfer of energy

Current in discrete path has both *numerical value and direction*; it is *measure of the rate at which charge is moving past a given reference point in a specified direction*



$$i = \frac{dq}{dt}$$



I – constant current; **i** – instantaneous current

Current

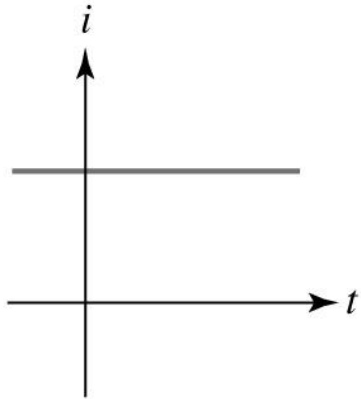
Charge transfer between time t_0 and t

$$\int_{q(t_0)}^{q(t)} dq = \int_{t_0}^t i dt$$

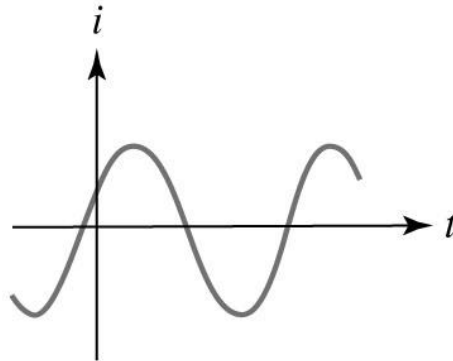
Total charge transfer over all time

$$q(t) = \int_{t_0}^t i dt + q(t_0)$$

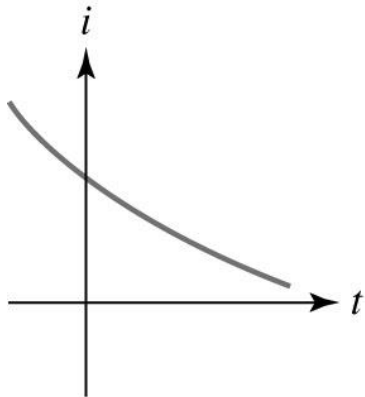
Current



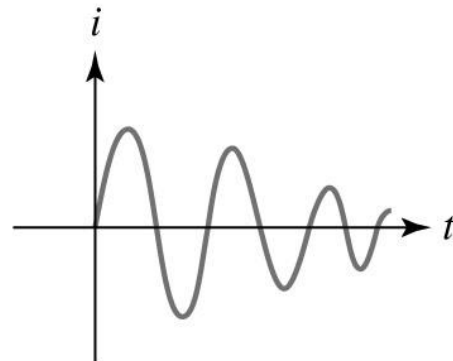
(a)



(b)



(c)



(d)

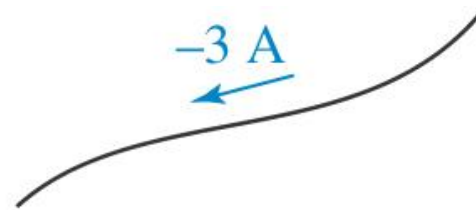
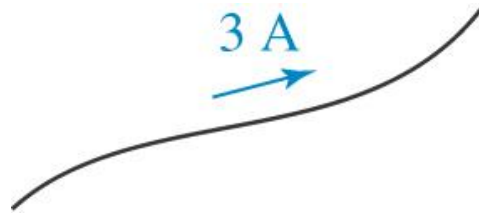
a – direct current (DC)

b – sinusoidal current

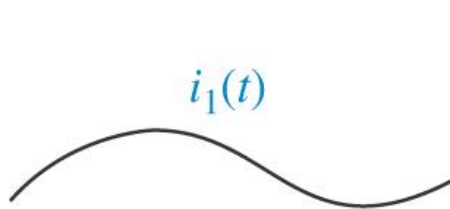
c – exponential current

d – damped sinusoidal
current

Current



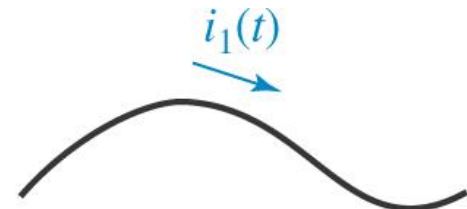
Two different methods of labeling the same current.



(a)



(b)

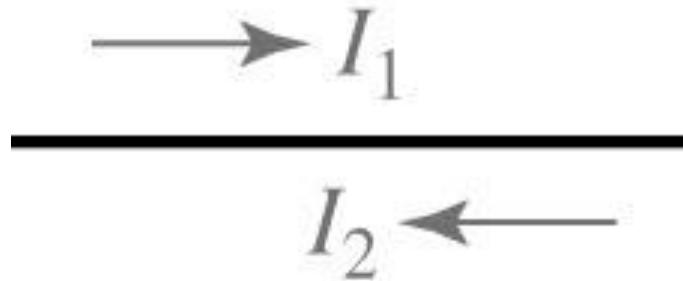


(c)

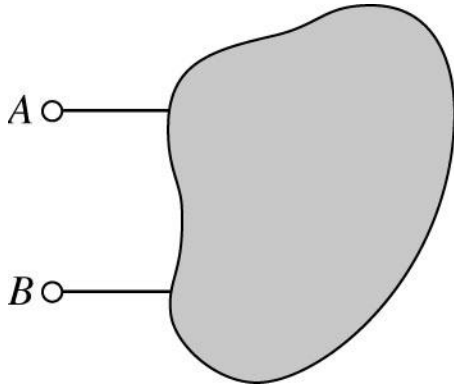
(a,b) Incomplete, improper, and incorrect definitions of a current.
(c) the correct definition of $i_1(t)$.

Practice

In the wire electrons are moving *left to right* to create a current of 1 mA. Determine I_1 and I_2 .



Voltage



The electric potential difference ΔV between points A and B is defined as the change in the potential energy (final value minus initial value) of a charge q when it moves from A to B, divided by the size of the charge.

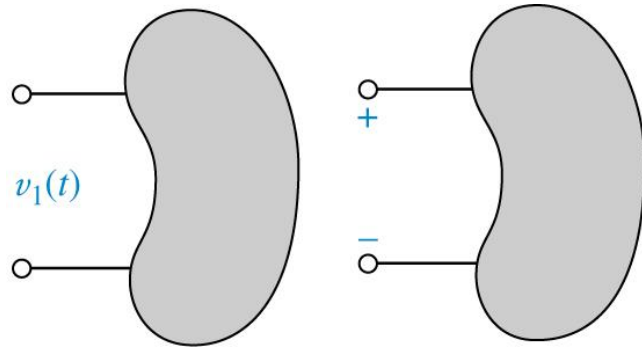
$$\Delta V = V_B - V_A = \Delta PE / q$$

The voltage across a terminal pair is a measure of the work required to move charge through the element

$$1 \text{ V} = 1 \text{ J/C}$$

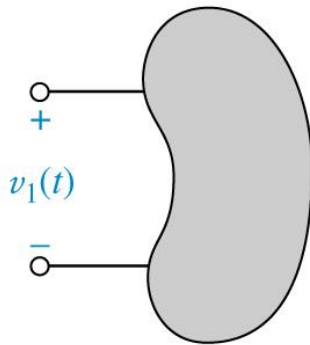
A voltage can exist between a pair of electrical terminals whether a current is flowing or not

DC Voltage



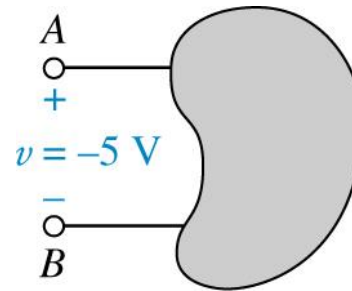
(a)

(b)

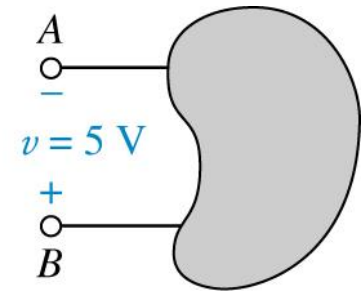


(c)

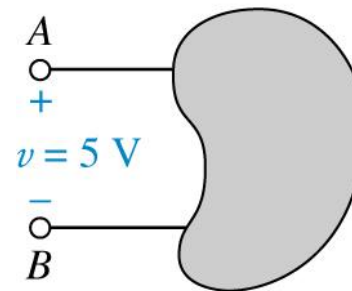
(a, b) These are inadequate definitions of a voltage. (c) A correct definition includes both a symbol for the variable and a plus-minus symbol pair.



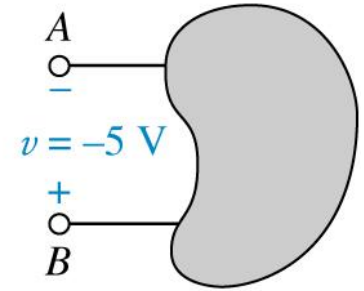
(a)



(b)



(c)

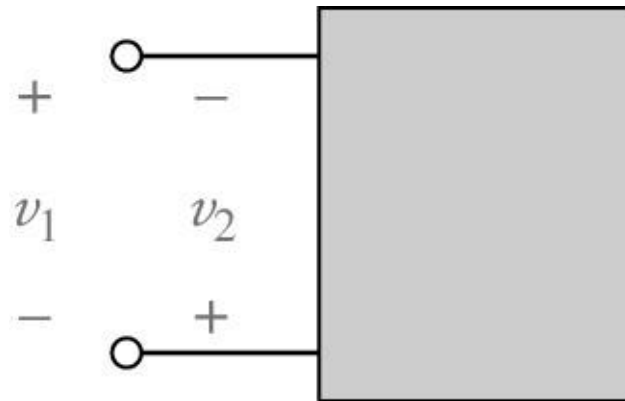


(d)

(a, b) Terminal B is 5 V positive with respect to terminal A; (c,d) terminal A is 5 V positive with respect to terminal B.

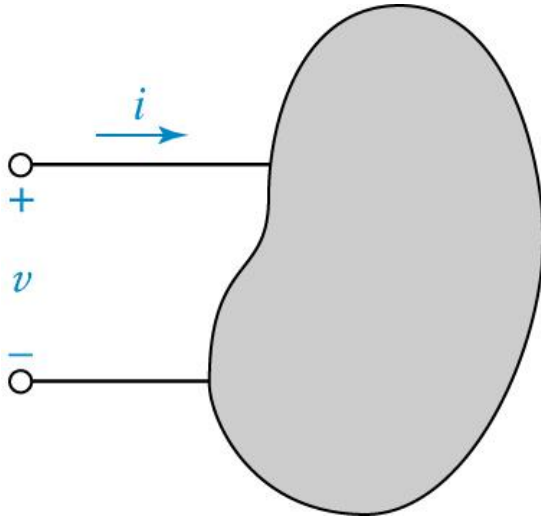
Practice

1. For the element, $v_1 = 17$ V. Determine v_2 .



Power

Power – the rate at which work is done or energy is expended



The power absorbed by the element is given by the product $p = vi$.

Alternatively, we can say that the element generates or supplies a power $-vi$.

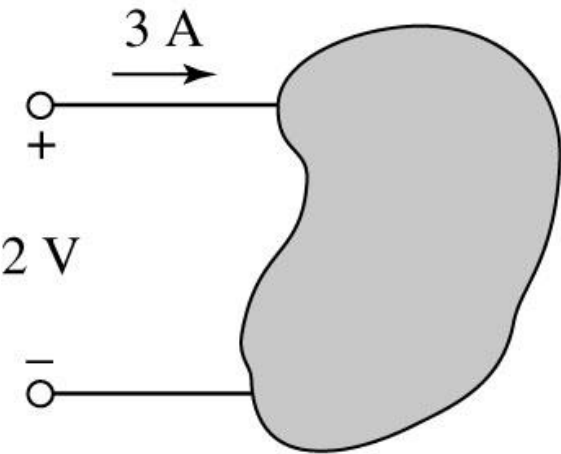
Passive sign convention!

If the current arrow is directed **into** the “+” marked terminal of an element, then **positive** $p = vi$ yields the **absorbed** power.

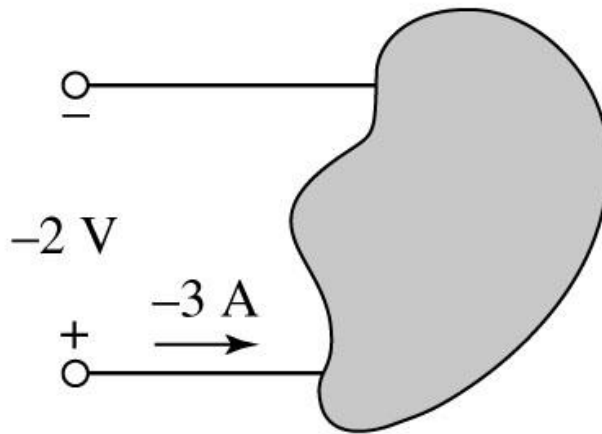
A negative value indicates that power is actually being generated by the element.

Example

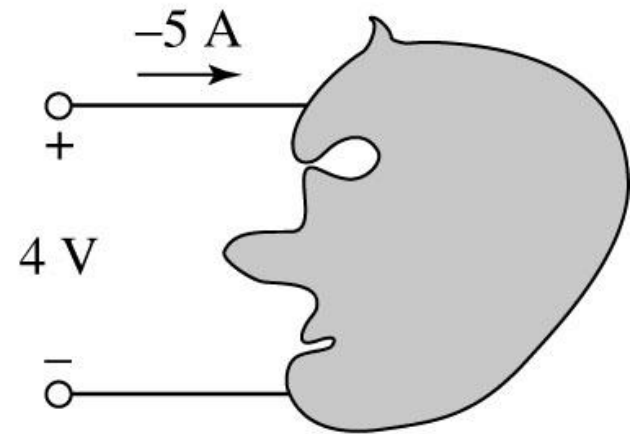
Compute the power absorbed by each part in the Fig.



(a)



(b)



(c)

Absorbing power:

$$a) P = (2 \text{ V})(3 \text{ A}) = 6 \text{ W}$$

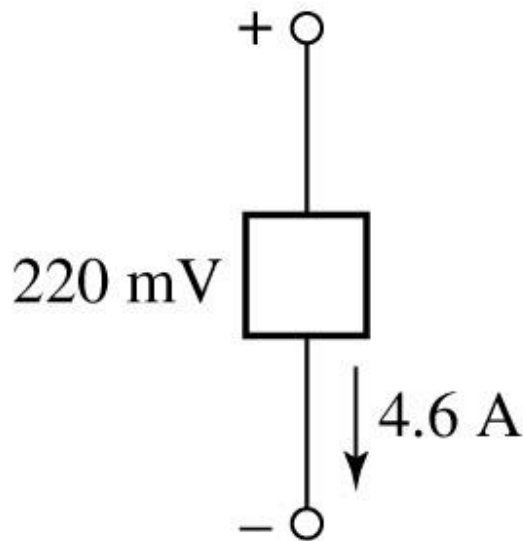
$$b) P = (-2 \text{ V})(-3 \text{ A}) = 6 \text{ W}$$

Supplying power:

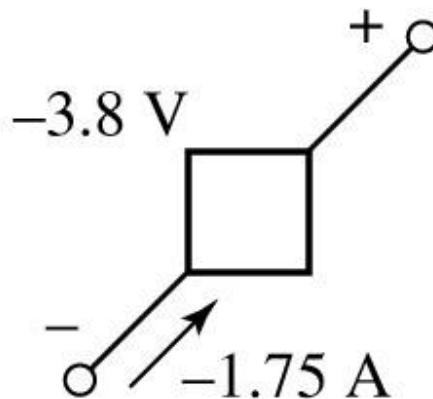
$$c) P = (4 \text{ V})(-5 \text{ A}) = -20 \text{ W}$$

Power

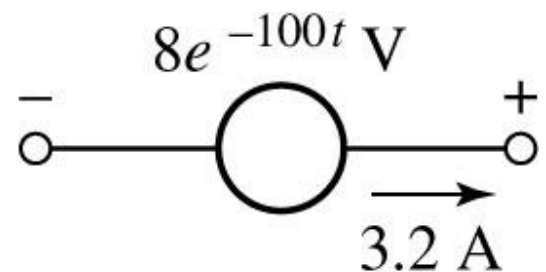
Practice



(a)



(b)

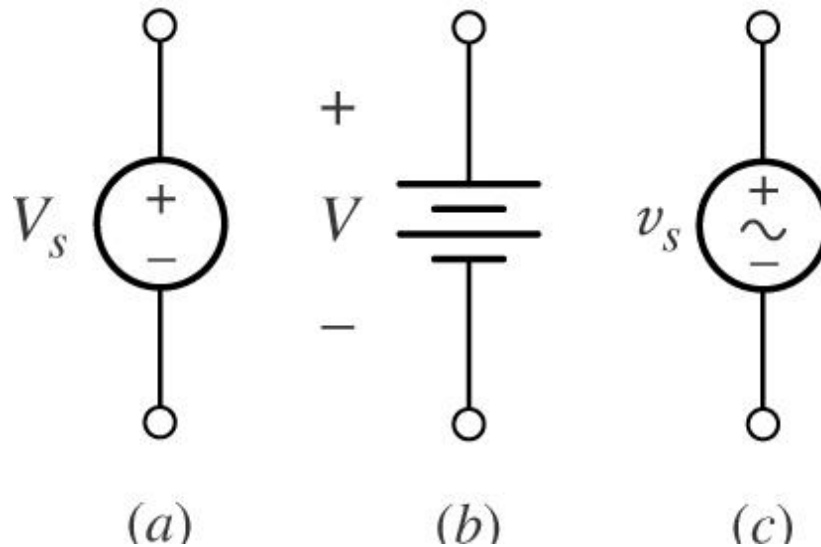


(c)

- Find the power being absorbed by the circuit element in Fig. a.
- Find the power being generated by the circuit element in Fig. b.
- Find the power being delivered to the circuit element in Fig. c at $t = 5 \text{ ms}$

Voltage & Current Sources

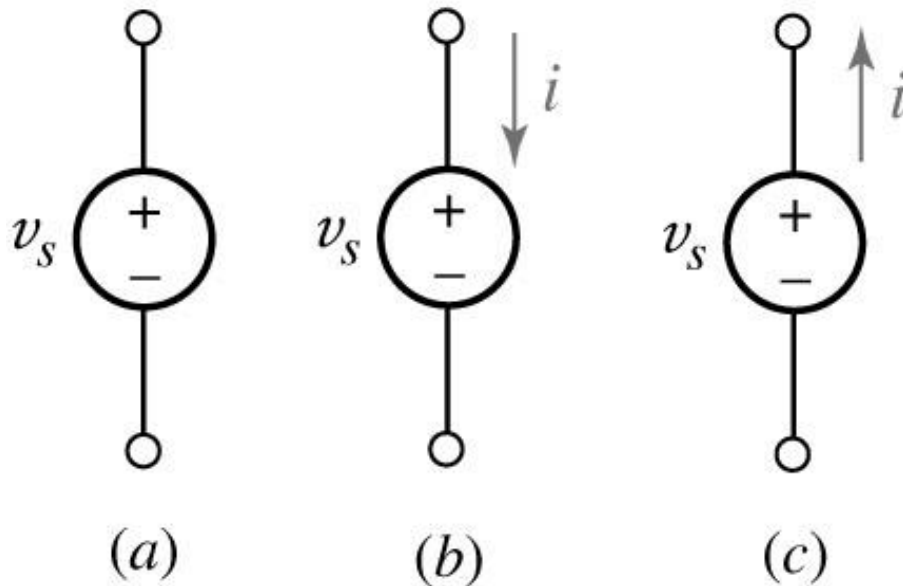
Independent Voltage Sources



Symbol for: (a) DC voltage source;
(b) battery; (c) AC voltage source.

Voltage & Current Sources

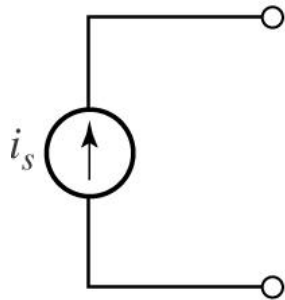
Independent Voltage Sources



Circuit symbol of the independent voltage source

Voltage & Current Sources

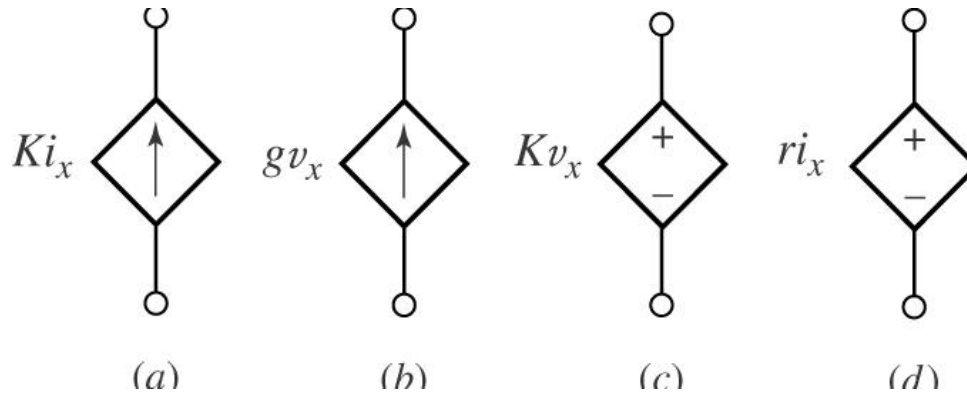
Independent Current Sources



Symbol for an independent current source.

Voltage & Current Sources

Dependent Sources



The four different types of dependent sources:

- (a) current-controlled current source;
- (b) voltage-controlled current source;
- (c) voltage-controlled voltage source;
- (d) current-controlled voltage source.

Scaling factors:

K – dimensionless

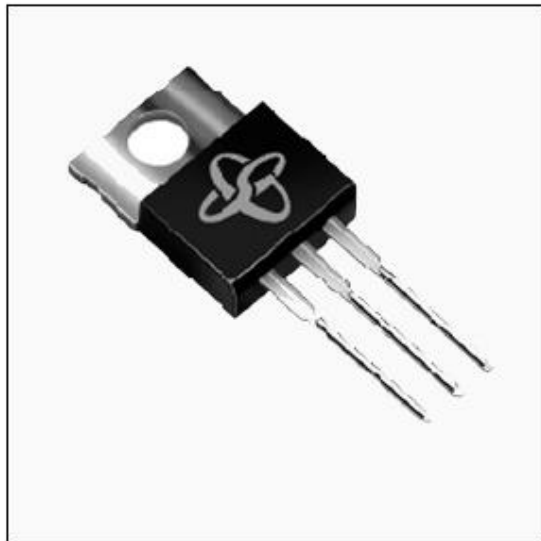
g – A/V

r – V/A

i_x and v_x – must be defined in the circuit

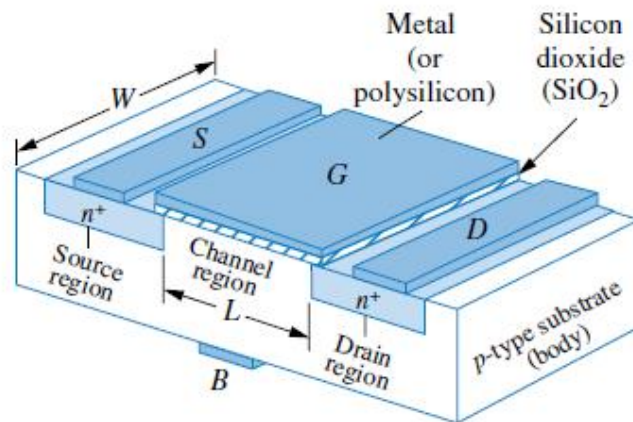
Networks & Circuits

Figure 2.22

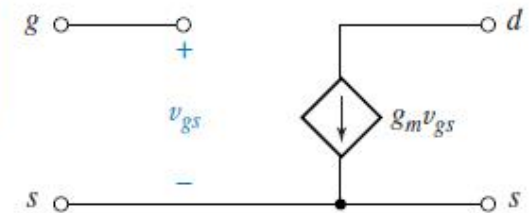


Courtesy of General Semiconductor

(a)



(b)



(c)

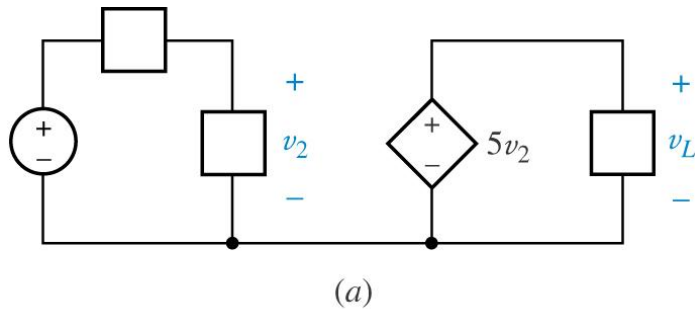
The Metal Oxide Semiconductor Field Effect Transistor (MOSFET). (a) discrete transistor in TO-5 package; (b) cross-sectional view (*R. Jaeger, Microelectronic Design, McGraw-Hill, 1997*); (c) equivalent circuit model for use in ac circuit analysis.

g_m - transconductance

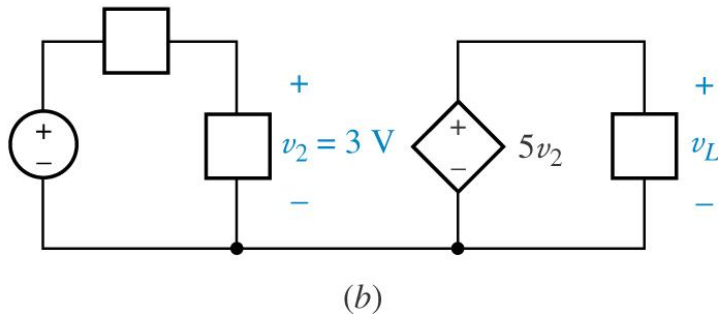
Voltage & Current Sources

Example

In the circuit below, if v_2 is known to be 3 V, find v_L .



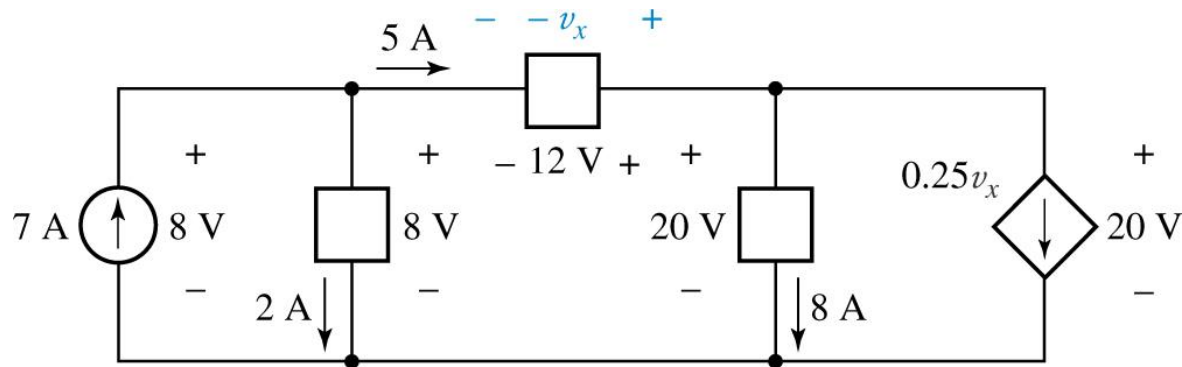
$$v_L = 5v_2$$



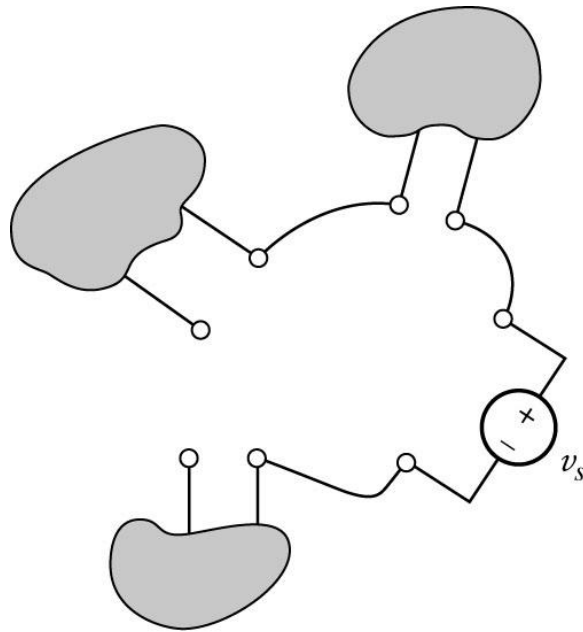
Voltage & Current Sources

Practice

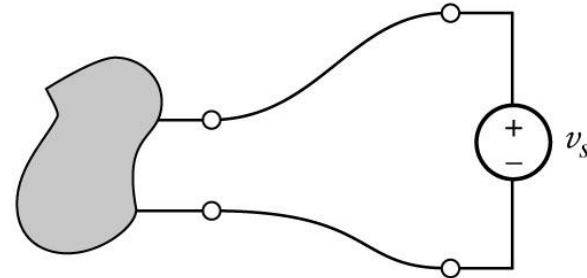
Find the power absorbed by each element in the circuit below.



Networks & Circuits



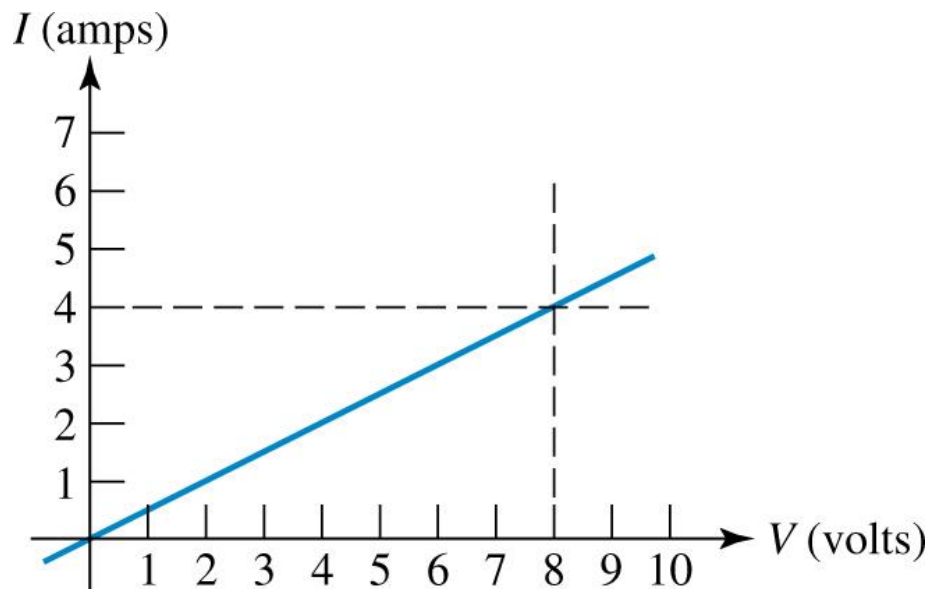
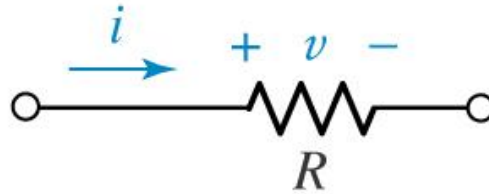
(a)



(b)

(a) A network that is not a circuit. (b) A network that is a circuit.

Ohm's Law

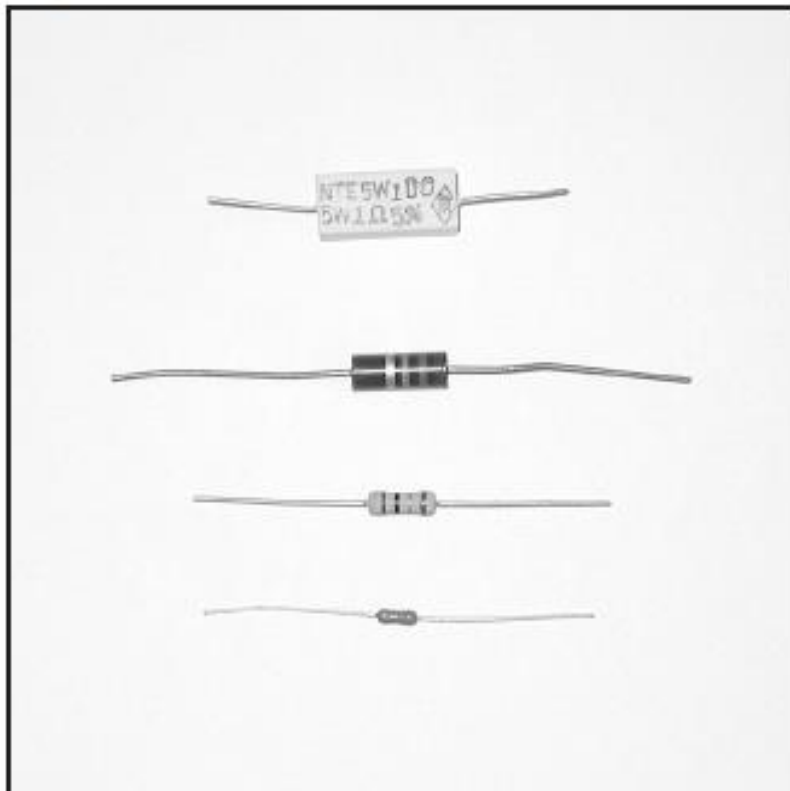


$$v = i R$$

or

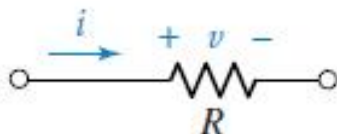
$$i = \frac{v}{R}$$

Power Absorption



$$p = vi = i^2 R = v^2 / R$$

(a)



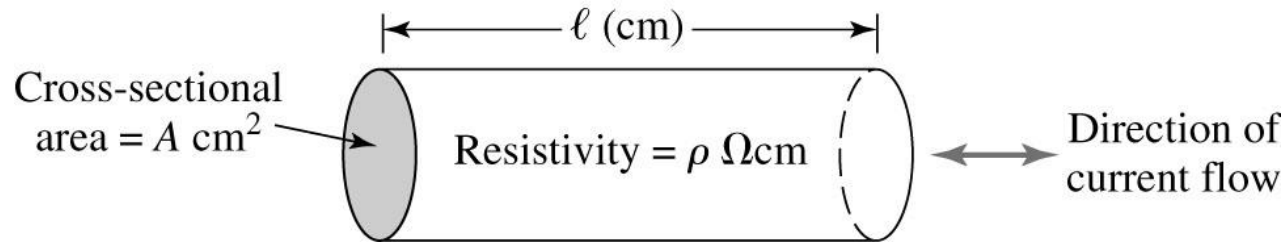
(b)

(a) Some representative resistors (larger packages are used for applications requiring significant power dissipation); (b) circuit symbol for the resistor.

Practice

A typical incandescent reading lamp runs at 60 W. If it is left on constantly, how much energy (J) is consumed per day and what is the weekly cost if energy is charged at a rate of 12.5 cents per kWh.

Wire Gauge



$$R = \rho \frac{l}{A}$$

$$\rho - \rho_0 = \rho_0 \alpha (T - T_0).$$

AWG – American Wire Gauge

Table 2.3 | Some common wire gauges and the resistance of (soft) solid copper wire.*

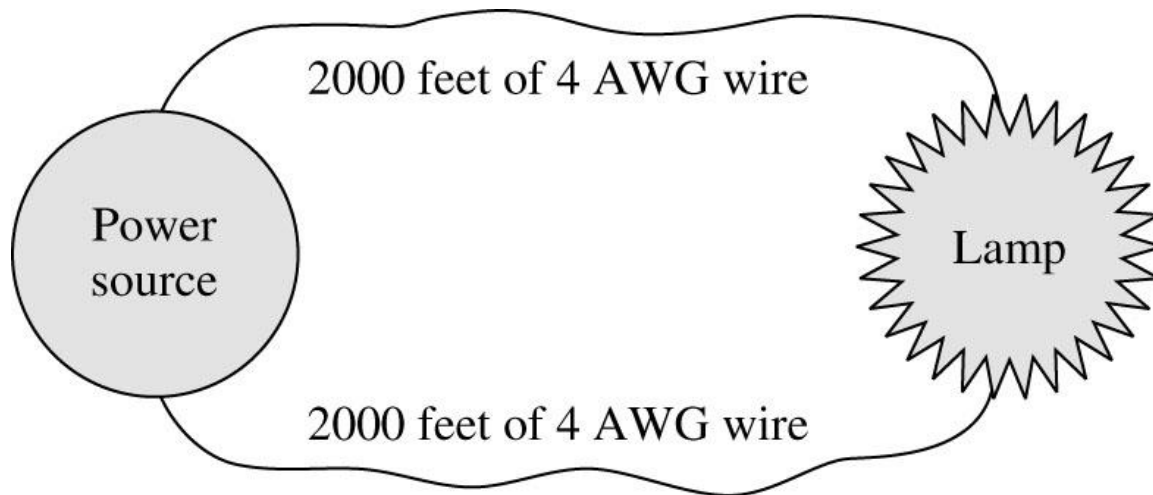
Conductor size (AWG)	Cross-sectional area (mm ²)	Ohms per 1000 ft at 20°C
28	0.0804	65.3
24	0.205	25.7
22	0.324	16.2
18	0.823	6.39
14	2.08	2.52
12	3.31	1.59
6	13.3	0.3952
4	21.1	0.2485
2	33.6	0.1563

*D. G. Fink and H. W. Beaty, *Standard Handbook for Electrical Engineers*, 13th Edition, McGraw-Hill, 1993, p. 4-47.

Wire Gauge

Example

A wire is run across a 2000-ft span to a high-power lamp that draws 100 A. If 4 AWG wire is used, how much power is dissipated (i.e., lost or wasted) within the wire?



$$R = (4000 \text{ ft})(0.2485\Omega / 1000 \text{ ft}) = 0.994\Omega$$

$$p = i^2R = (100 \text{ A})^2(0.994 \Omega) = 9.94 \text{ kW dissipated by the wire}$$

Practice

Faced with the significant power losses described in the previous example, your manager instructs you to have the 4 AWG wire replaced with 2AWG. Calculate the power lost in the new wire, assuming the lamp is still drawing 100 A. Out of curiosity, how much more will the new wiring weigh (two times more, four times more, etc.)?

Conductance

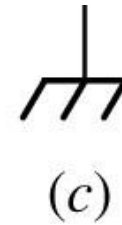
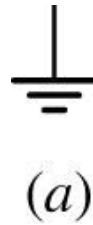
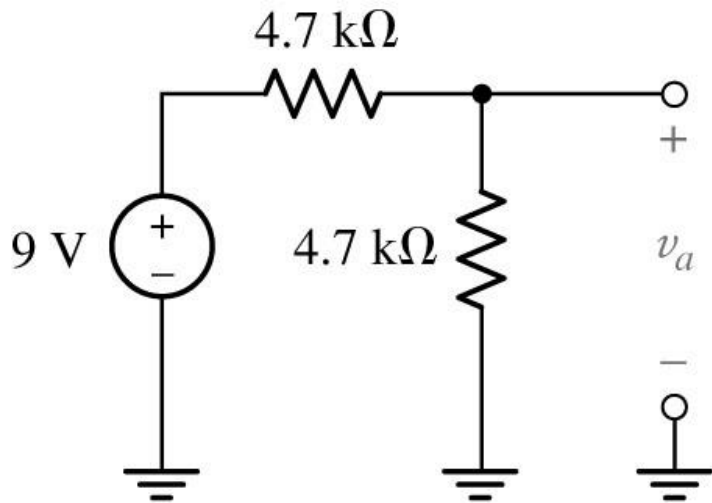
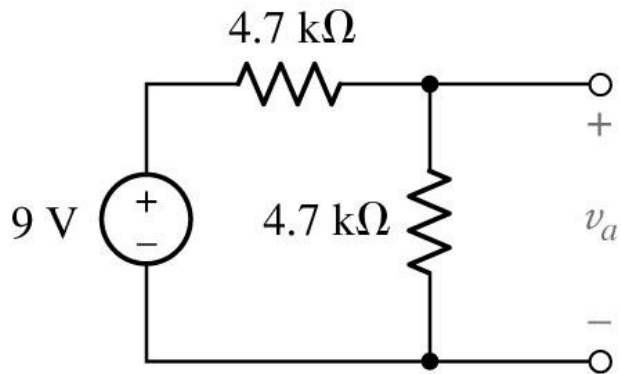
$$\frac{i}{v} = \frac{1}{R} = G \quad \text{– conductance (constant)}$$

SI units – Siemens (1S=1A/1V)

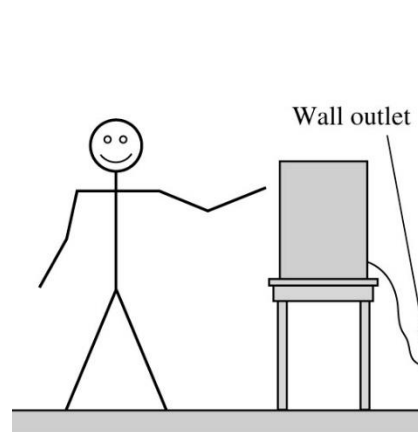
$$p = vi = v^2 G = \frac{i^2}{G}$$

$$\frac{1}{\rho} = \sigma \quad \text{– conductivity}$$

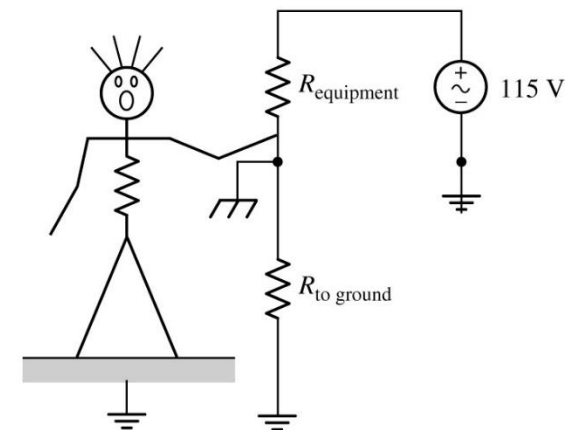
Grounding



- a) earth ground
- b) signal ground
- c) chassis ground



(a)



(b)

Diodes

<https://www.youtube.com/watch?v=gkjCe0r5-cw>

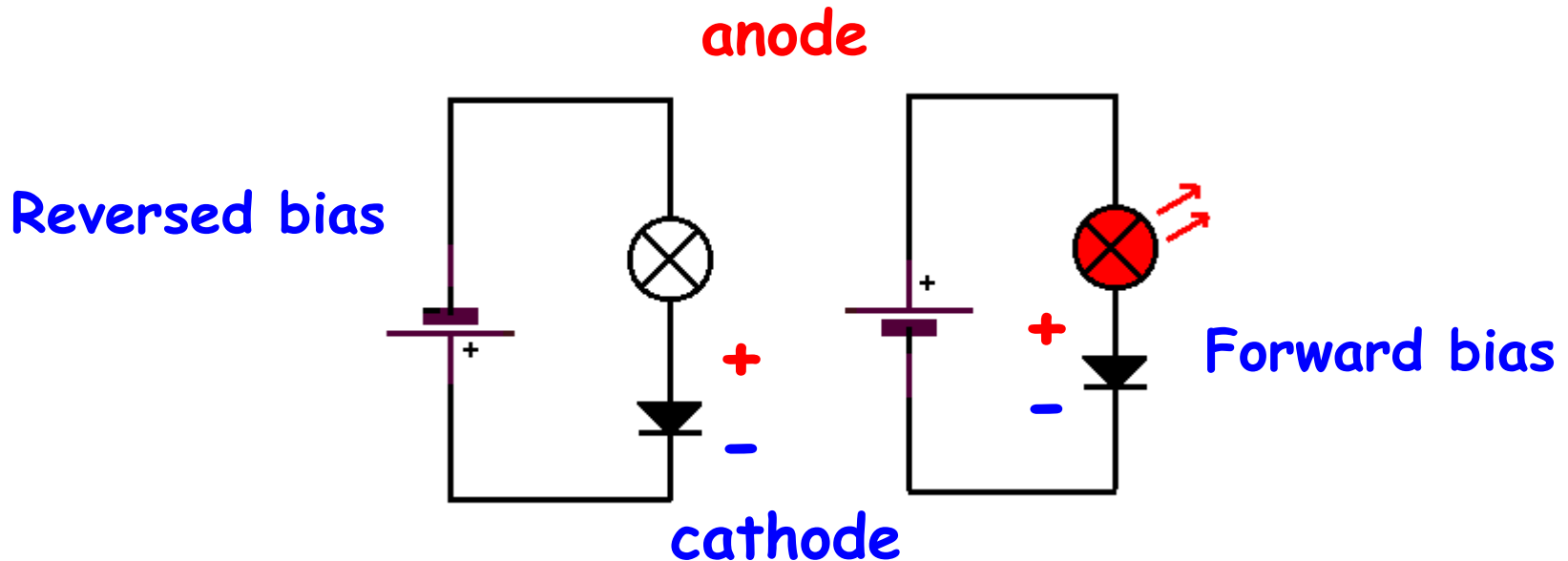
<https://www.youtube.com/watch?v=AgkQrCeJF1Y>

<https://www.youtube.com/watch?v=U8daujO20nM>

<https://www.youtube.com/watch?v=HzwGOZ1QBXU&list=PL9P80IAEKoGT7M1DyMo0K8FwhPbjQz7tt&index=3>

<https://www.youtube.com/watch?v=4SlfaocMfdA&list=PL9P80IAEKoGT7M1DyMo0K8FwhPbjQz7tt>

Practical aspects



The left hand diagram shows **reverse bias**, with positive on the cathode and negative on the anode (via the lamp). **No current flows.**

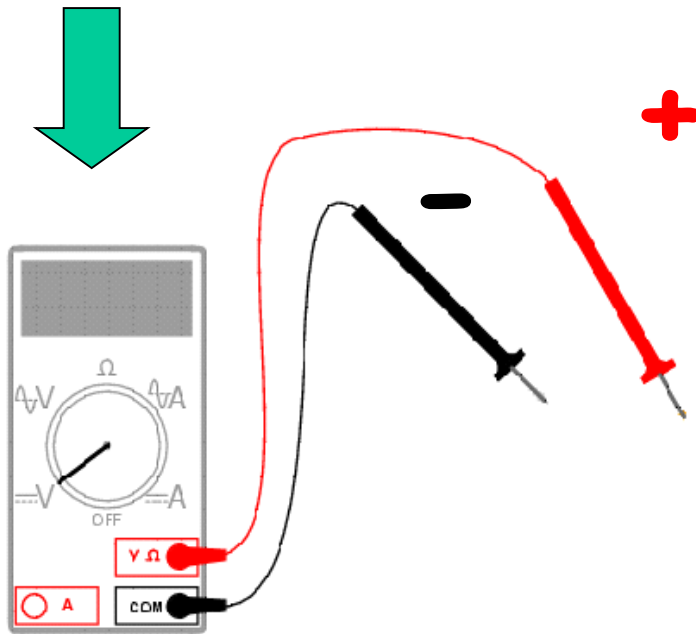
The other diagram shows **forward bias**, with positive on the anode and negative on the cathode. **A current flows.**

Diodes

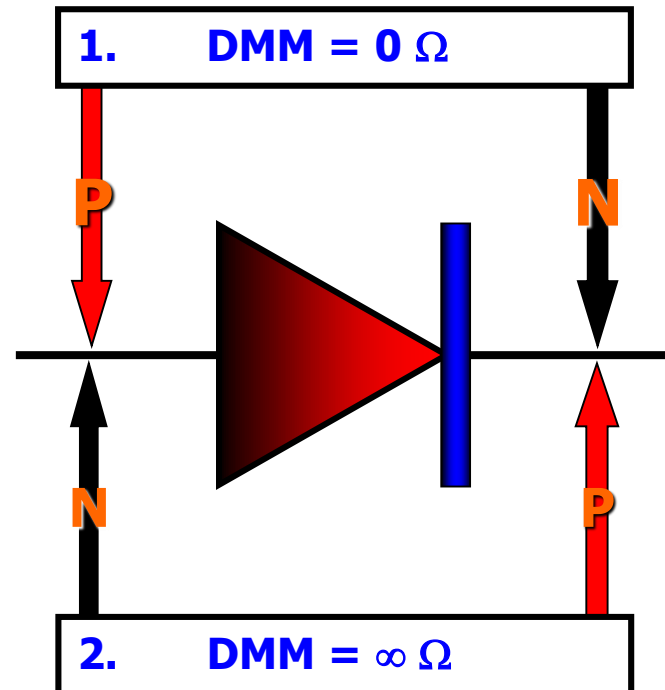
Practical aspects

Diode Ohms Check:

Check by reversing the leads on the Digital Voltage Mutimeter (DMM).



DMM

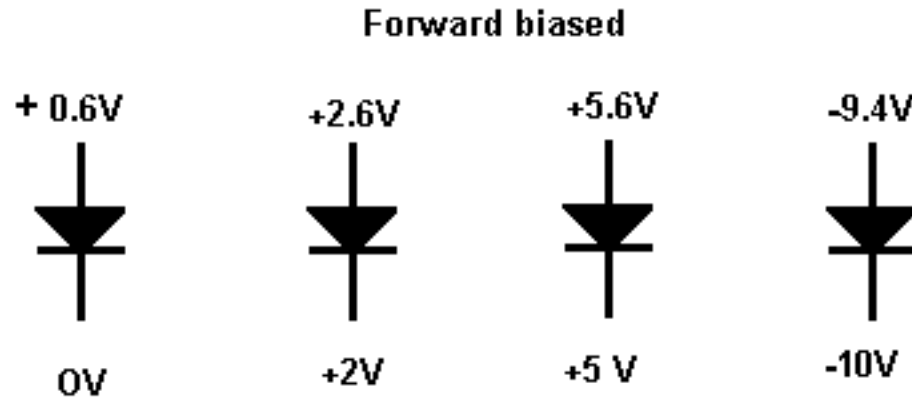


Diodes

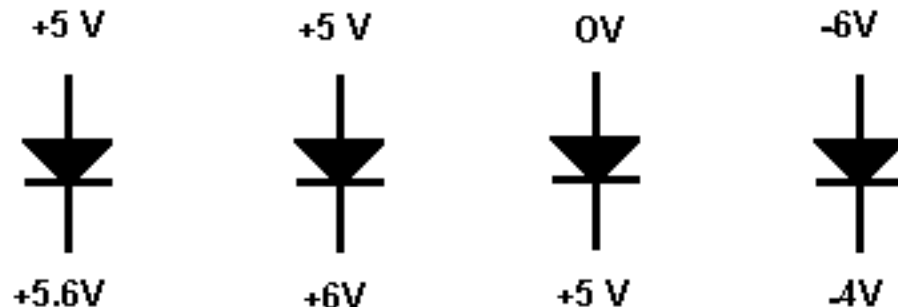
Practical aspects

Diode Voltages

To forward bias a diode, the anode must be more positive than the cathode or **LESS NEGATIVE**.



To reverse bias a diode, the anode must be less positive than the cathode or **MORE NEGATIVE**.

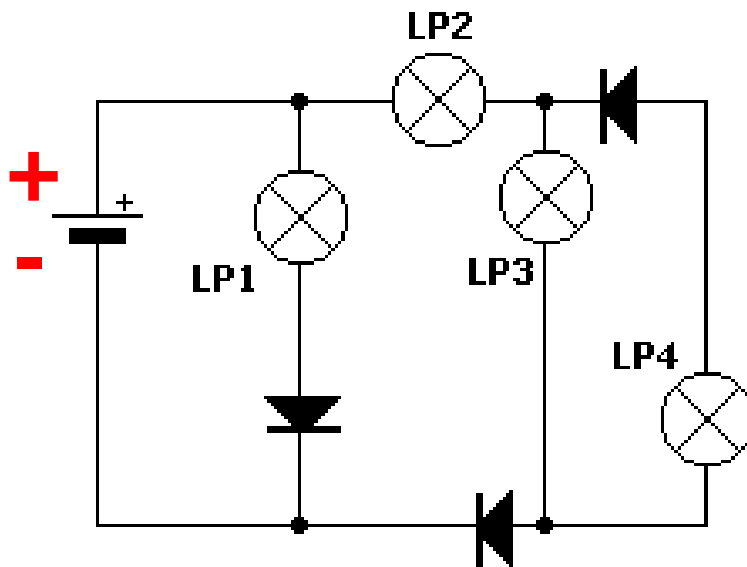


Reverse biased

A conducting diode has about 0.6 volts across if silicon, 0.3 volts if germanium.

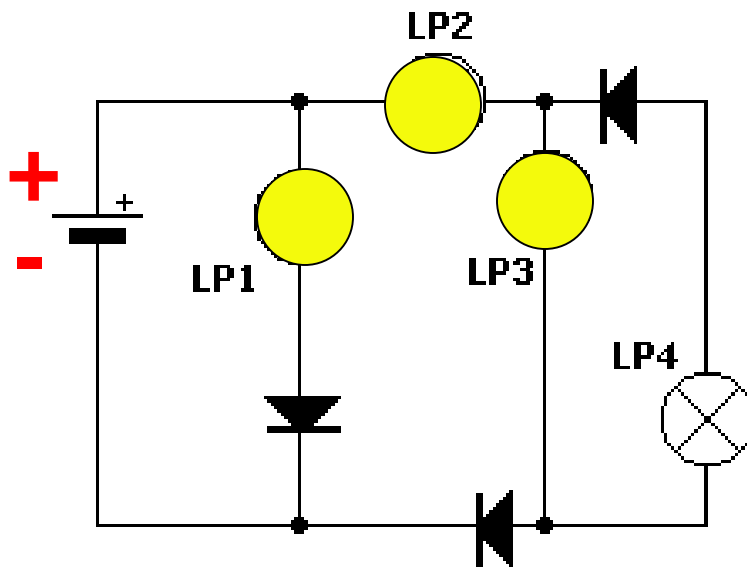
A Diode Puzzle

Which lamps are alight? Some may not be full brightness.



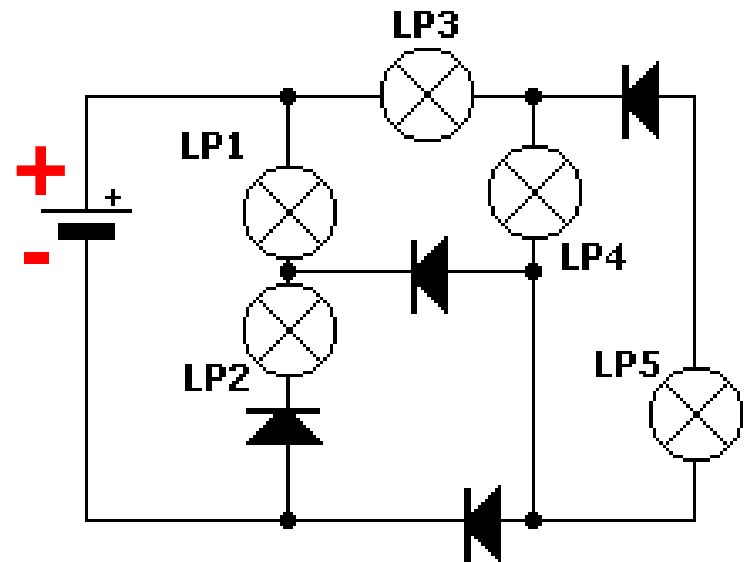
A Diode Puzzle

Which lamps are alight? Some may not be full brightness.



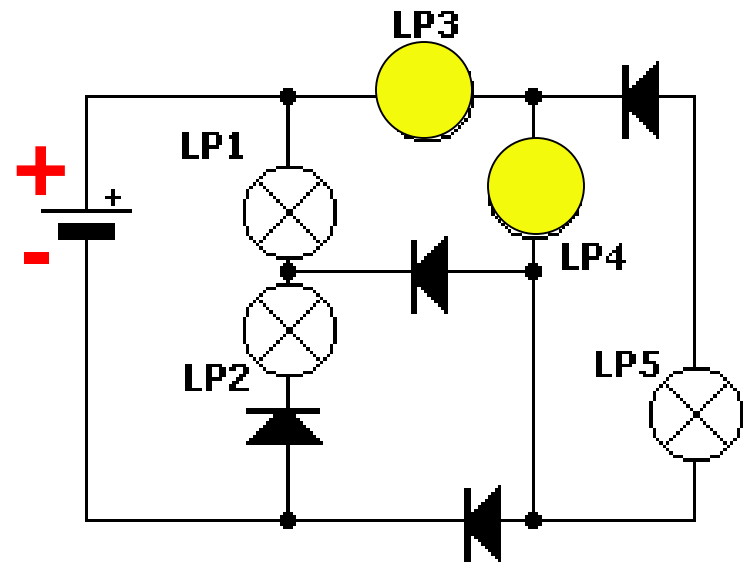
Exercise - a Diode Puzzle

Which lamps are alight? Some may not be full brightness.

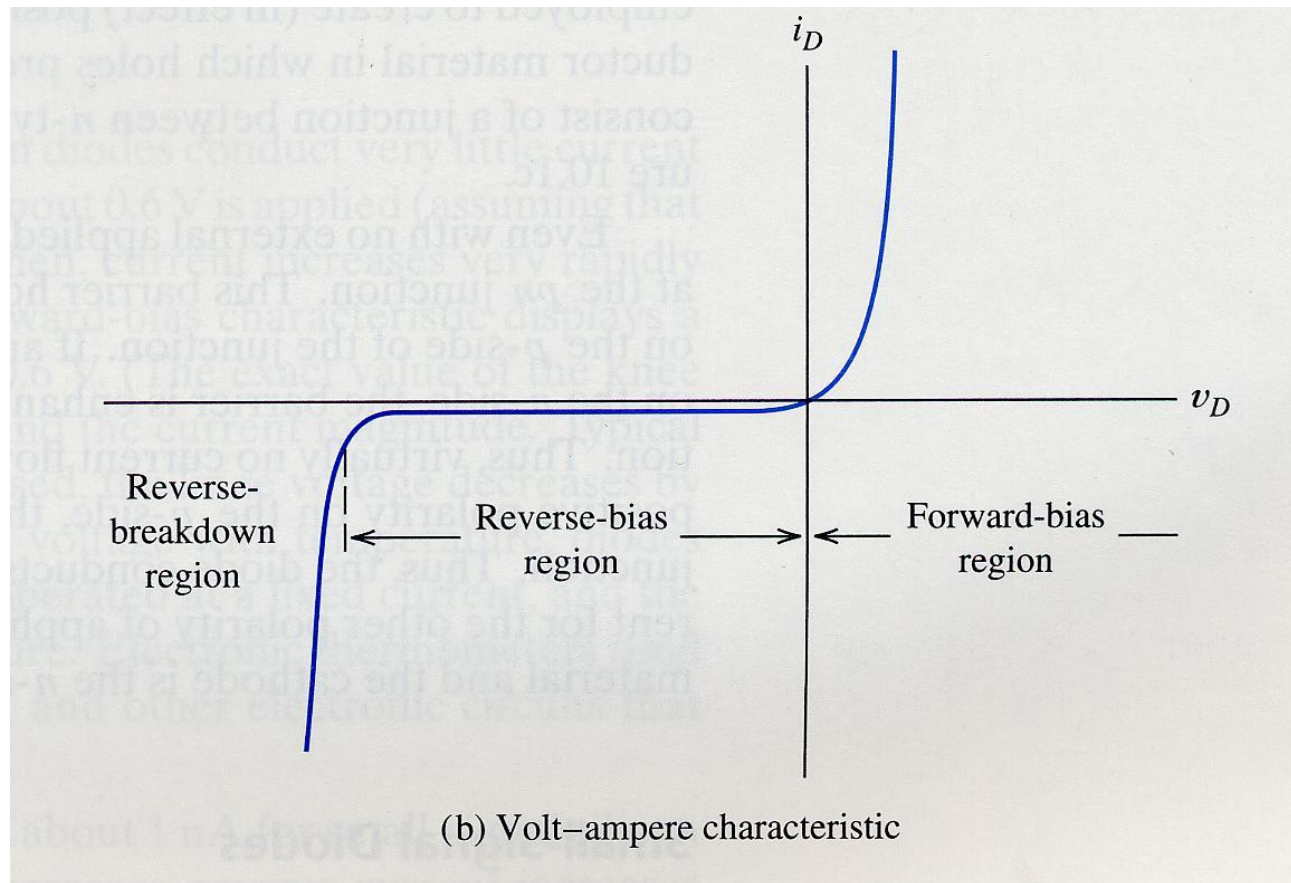


Exercise - a Diode Puzzle

Which lamps are alight? Some may not be full brightness.

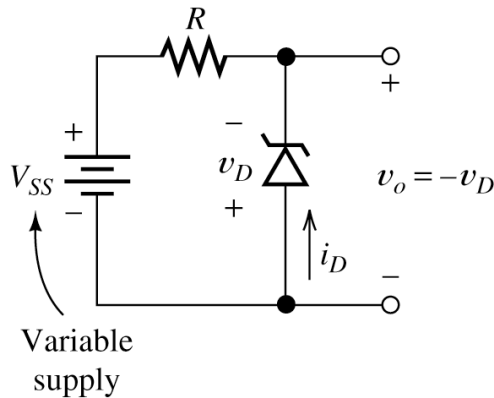


Practical aspects

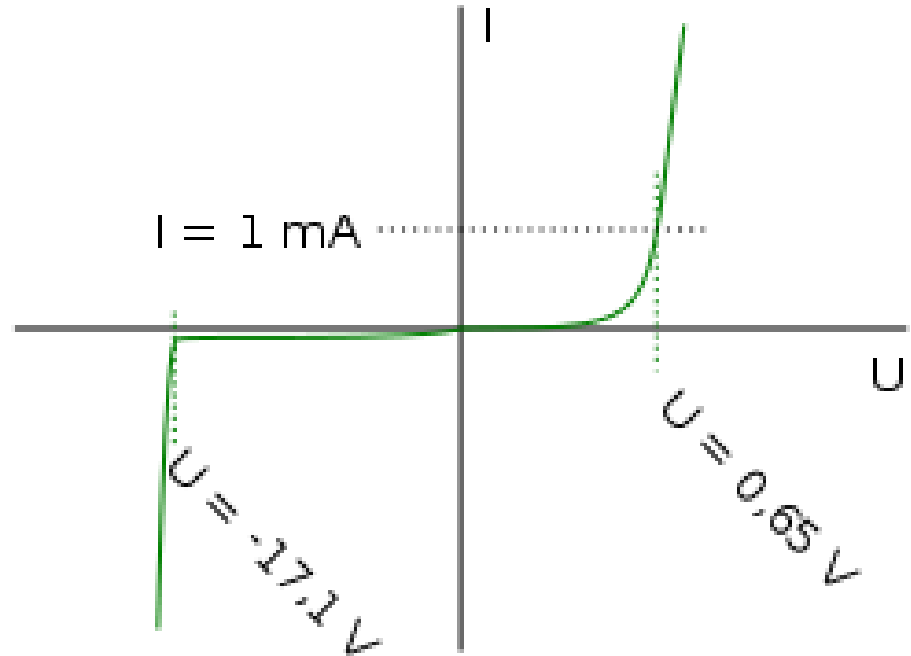


A diode is a nonlinear device and typical linear circuit analysis methods do not apply!

Zener Diode - Voltage Regulator (reverse biased)



A simple regulator circuit that provides a nearly constant output voltage v_o from a variable supply voltage.



A Zener diode is a type of diode that permits current not only in the forward direction like a normal diode, but also in the reverse direction if the voltage is larger than the breakdown voltage known as "Zener knee voltage" or "Zener voltage".



SUMMARY to Chapter 2

1. The system of units most commonly used in electrical engineering is the SI
2. The direction in which positive charges are moving is the direction of positive current flow; alternatively, positive current flow is in the direction opposite that of moving electrons
3. To define a current, both a value and a direction must be given. Currents are typically denoted by the uppercase letter “I” for constant (dc) values, and either $i(t)$ or simply i otherwise
4. To define a voltage across an element, it is necessary to label the terminals with “+” and “–” signs as well as to provide a value (either an algebraic symbol or a numerical value)
5. Any element is said to supply positive power if positive current flows out of the positive voltage terminal. Any element absorbs positive power if positive current flows into the positive voltage terminal

SUMMARY to Chapter 2

6. There are six sources: the independent voltage source, the independent current source, the current-controlled dependent current source, the voltage-controlled dependent current source, the voltage-controlled dependent voltage source, and the current-controlled dependent voltage source
7. Ohm's law states that the voltage across a linear resistor is directly proportional to the current flowing through it; i.e., $v = i R$
8. The power dissipated by a resistor (which leads to the production of heat) is given by $p = vi = i^2 R = v^2 / R$
9. Wires are typically assumed to have zero resistance in circuit analysis. When selecting a wire gauge for a specific application, however, local electrical and fire codes must be consulted